Hair Cutting Robots

Lecture 2: Rust and Simulator Setup

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1 Installing Rust

What is Rust

A Compiler Language

Rust is a programming language which have both lower-level language's efficacy and higher-level language's feature.

GC? Ownership!

Different from lower-level language we need to manually operate on memory which causes potential bugs or like higher-level language using GC (Garbage collection) which contributes to runtime-overhead, rust introduce an ownership model to make sure programs are safe and fast.

Introduced in Linux

Rust is introduced as Linux's important component. Such as SudoRS, KernelRS.
`https://github.com/trifectatechfoundation/sudo-rs`
`https://docs.kernel.org/rust/index.html`

Why we learn Rust

Compiler is REALLY strong

Compilers can identify most of errors but not leaving them to runtime. It can detect the following errors:

- Unsafe memory usage (such as double free)
- Accessing undefined fields/functions
- And so on

Zero Runtime Overhead

Compiled in advance, what you write is what the program cost.
No hidden usage (Such as background GC).

Emerging chance

Rust is introduced as Linux as discussed before.

Compiler Detected Errors

Mutable and Not-accessed value warning

```
let mut a: i32 = 5;     // value assigned to ‘a‘ is never read - maybe overwritten
a = 10;
println!("The value of a is: {}", a);

let a: i32 = 15;        // value assigned to ‘a‘ is never read - maybe overwritten
a = 10;                 // cannot assign twice to immutable variable ‘a‘
println!("The value of a is: {}", a);
```

Type checking

```
let a: String = String::from("Some String Pointer");
a = "Another String Pointer";   // mismatched types - expected ‘String‘
println!("The value of a is: {}", a);
```

RustUp

What is RustUp

The official Rust toolchain installer and version manager. It installs and manages multiple Rust versions and associated tools.

What you get

- `rustc` — the Rust compiler
- `cargo` — package manager & build tool
- `rustfmt` — code formatter
- `clippy` — linter

Install

MacOS / Linux

1. Run `curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh`
2. Restart your terminal or run: `source $HOME/.cargo/env`

Windows

1. Download RustUp installer on `rustup.rs`
2. Double click and follow the prompts.
3. If you don't have C++ environment, you may need them from `visualstudio.microsoft.com`

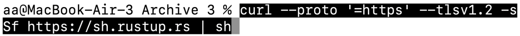
MacOS CLI

Open the Terminal APP

If you are using MAC, press `Cmd + Space` and type "Terminal". Then press `Enter` to open the terminal. (See the figure left below)

Typing the Command

When you go into the terminal, you will see a command prompt. Click on the window and type the exact command provided. Then press `Enter` to execute the command. (See the figure right below)



Setting PATH on Linux / MacOS

If `rustc` is not found after installation, the `cargo` `bin` directory is not on your `PATH` yet.

Quick check

```
echo $PATH
ls $HOME/.cargo/bin
```

Add it for the current shell

```
source $HOME/.cargo/env
```

Add it permanently

Append the following line to your shell config file (`~/.zshrc` on macOS, `~/.bashrc` on most Linux), then restart the terminal or run `source` on that file:

```
export PATH="$HOME/.cargo/bin:$PATH"
```

Verify

In CLI check

`rustc --version`, If you install successfully you will see the output of the current rust version.

If you are not successful, you may need to check whether the path environment is correctly set.

Create a new Rust Project!

Cargo is one component from the Rust toolchain. Run `cargo new <project_name>` to create a project!

Windows? Switch to WSL

Supported environments
In this course we only help with problems on **macOS** and **Debian-based Linux** (Debian / Ubuntu).

If you are on Windows
If anything breaks on native Windows, please switch to **WSL** (Windows Subsystem for Linux) and continue from there.

Native Windows issues are your own responsibility to fix.

Installing WSL

Install WSL (Ubuntu)

Open PowerShell as Administrator and run:

```
wsl --install -d Ubuntu
```

After installation

Reopen the terminal, launch Ubuntu, and redo the Rust installation steps inside the Linux shell.

Using WSL for Webpages

Visiting webpages from WSL
Use `curl` from **inside the Linux shell**, not from Windows PowerShell or CMD.

```
curl https://sh.rustup.rs
```

This ensures the request is made from the Linux environment, so downloaded files and installers land in the Linux file system where Rust will run.

2 Get Started with Rust